



DockM8 User Guide

Introduction

This is a guide to the usage of the DockM8 v1.1.0 package.

If you experience any issues or bugs, please report them using the issue-reporting system on GitHub: [GitHub Issues](#). Please use appropriate tags (e.g., 'bug', 'installation') to help us categorize and address your issue.

1 Usage

1.1 Graphical User Interface

To use the Graphical User Interface, follow these steps:

- Open Terminal (Ctrl + Alt + T on Ubuntu, or launch the Ubuntu app on Windows)
- Navigate to the DockM8 folder:

```
cd ~/DockM8
```

- Activate the dockm8 conda environment:

```
conda activate dockm8
```

- Start the GUI by running the following command:

```
streamlit run gui.py
```

- A link will appear in the command's output which contains localhost, click on it.
- The GUI will now start in your browser.



Setup

Which mode do you want to run DockM8 in? ?

single ▼

Number of CPUs ?

1

28

32

Choose a software directory ?

/home/tony/DockM8_v1/software

- **Setup:**

- Mode: Choose which mode you want to run DockM8 in. Single mode docks the compound library to a single receptor/protein. Ensemble mode docks the compounds to several receptors and outputs the best-scoring compounds across all receptors. When ensemble mode is selected, an additional “Ensemble Top Percent” slider appears to control the percentage of top-ranked compounds reported across all receptor structures.
- CPUs: Choose the number of CPUs you want to run the workflow with. This defaults to the number of CPUs on your machine -10%.
- Software: Choose the directory where the software required for DockM8 has been installed. This defaults to the directory where DockM8 is installed.

☒ Generate decoys ?

Decoy generation

Enter the path to the active ligands file (.sdf format) ?

/home/tony/DockM8_v1/test_data/CDK2_actives.sdf

Number of decoys ?

1

20

100

Which decoy generation model do you want to use? ?

DUD-E ▼

- **Decoy Generation:** DockM8 can generate decoys if provided with a list of active compounds. When the “Generate Decoys” option is enabled, the following inputs become available:

- Actives File: Select the path to the .sdf file containing the active compounds. Compounds do not need to be labelled, DockM8 will take care of the labelling.
- Number of Decoys: Set the number of decoys generated per active compound.
- Decoy Model: Choose the decoy generation model. Options are: DUDE, DEKOIS, or DUDE_P.

Note: decoy generation is not supported in ensemble mode.

Receptor(s)

Receptor 1

`/home/tony/DockM8_v1/test_data/4kd1_p.pdb`

Add Receptor

☒ Prepare receptor ?

Advanced Protein Preparation ^

Protonation method ?

protoss v

☒ Fix nonstandard residues ?

☒ Fix missing residues ?

☒ Remove heteroatoms ?

☒ Remove water molecules ?

• Receptor(s):

- Path: Enter the path of the .pdb file of the receptor/protein. Additional receptor paths can be added by clicking the **Add Receptor** button. Each extra receptor gets a corresponding **Remove** button to delete it. Ensemble mode requires at least two valid receptors.
- Prepare Receptor: When this option is enabled, the protein will be prepared for docking. The following preparation options become available:
 - * Protein Protonation: Choose the protonation method for the receptor. Options are: `protoss` (Protoss web-server), `pdbfixer` (PDBFixer library), or `none` (skip protonation).
 - * Fix Nonstandard Residues: Fix nonstandard residues during protein preparation (default: enabled).
 - * Fix Missing Residues: Fix missing residues during protein preparation (default: enabled).
 - * Remove Heteroatoms: Remove heteroatoms from the receptor (default: enabled).
 - * Remove Water: Remove water molecules from the receptor (default: enabled).

Pocket finding

How should the pocket be defined? ?

Reference v

Pocket radius ?

10 - +

Reference Ligands

Reference Ligand 1 (for Receptor 1)

`/home/tony/DockM8_v1/test_data/4kd1_l.sdf`

- **Pocket Finding:** Choose how the binding pocket should be defined. The following options are available:
 - Reference: Uses a reference (co-crystallised) ligand to define the pocket. Requires a reference ligand file per receptor.
 - RoG: Uses a reference ligand's radius of gyration to define the pocket. Requires a reference ligand file per receptor.

- **Dogsitescorer:** DockM8 will use the DogSiteScorer webserver to determine pocket coordinates automatically. A “DogSiteScorer Method” selector becomes available with options: Volume (default), Druggability_Score, Surface, or Depth.
- **Custom:** Manually define the pocket center and size coordinates. Note: the Custom pocket definition is not currently supported in ensemble mode.
- **Pocket Radius:** When using Reference or RoG modes, a “Pocket radius” input appears where you can set the radius (in Ångstroms) used to define the binding pocket. Default=10.
- **Reference Ligands:** When using Reference or RoG modes, a “Reference Ligand” input appears for each receptor (labelled “Reference Ligand N (for Receptor N)”). Each reference ligand corresponds to its matching receptor in ensemble mode.

Ligands

Enter the path to the ligand library file (.sdf format) ?

/home/tony/DockM8_v1/test_data/library.sdf

Choose the column name that contains the ID of the ligand ?

ID

Ligand conformers

How should the conformers be generated? ?

MMFF ▼

Ligand protonation

How should the ligands be protonated? ?

GypsumDL ▼

Advanced Ligand Preparation ^

☒ Standardize compound IDs ?

☒ Remove salts ?

Minimum pH ?

6.40 - +

Maximum pH ?

8.40 - +

pKa precision ?

1.00 - +

- **Ligands:**
 - Path: Select the path of the .sdf file containing the library of ligands to dock to the receptor.
 - ID: If the ligands are not named in the .sdf file, or you want to use a specific column in the .sdf as the ligand name, you can set that here.
 - Conformers: Ligand conformers can be generated with RDKit using the UFF or MMFF forcefield, or with GypsumDL.
 - Protonation: Ligand protonation can be enabled by selecting the GypsumDL option. Selecting None will leave the protonation state as it is in the docking library .sdf file. When using GypsumDL, you can configure the pH range (min/max) and pKa precision.
 - Standardize IDs: Standardize compound identifiers (default: enabled). Disabling this may cause errors with numeric-only IDs or IDs containing special characters.

- Remove Salts: Remove salt counterions from ligands (default: enabled).

Docking programs

Choose the docking programs you want to use

GNINA x

Number of poses

10

1 100

Exhaustiveness

1 32

☐ Bust poses using PoseBusters: WARNING may take a long time to run

Pose Selection

Choose the pose selection methods you want to use

bestpose x

Scoring functions

Choose the scoring functions you want to use

GNINA-Affinity x CNN-Score x

Consensus

Choose which consensus algorithm to use:

rbr

- **Docking Programs:**

- Programs: Select one or multiple docking programs to perform the docking step. Available programs: GNINA, GNINA_GPU (GPU-accelerated), SMINA, QVINA2, QVINAW, PLANTS. Ensure the docking programs are suitably installed and you have permission to run them.
- Number of poses: Select the number of poses to generate for each compound in the library. This setting defaults to 10.
- Exhaustiveness: For GNINA, GNINA_GPU, SMINA, QVINA2 and QVINAW, the exhaustiveness (or precision) of the docking step can be defined. Note that this increases the run time.
- Pose Busting: The checkbox controls whether the pose quality control using PoseBusters is enabled. Note that this may significantly increase the run time.

- **Pose Selection:** Here you can choose one or more Pose Selection methods. The bestpose method selects the best pose from each docking program based on the docking score. Alternatively, you can use any of the available rescoring functions (see Scoring Functions below) to select the best pose for each compound according to the specified scoring function.
- **Scoring Functions:** Here you can choose which scoring functions are used in the consensus score calculation. Available functions: AD4, CHEMPLP, CNN-Affinity, CNN-Score, ConvexPLR, GenScore-balanced, GenScore-docking, GenScore-scoring, GNINA-Affinity, GNINA-All, KORP-PL, LinF9, NNScore, PLP, RFScoreVS, RTMScore, SCORCH, Vinardo. Multiple choices are required for the consensus option to be available.
- **Consensus:** Here you can choose which consensus algorithm you want to use for the final scoring of the compound library. Available methods: ecr (Element Count Rank), rbr (Rank-Based Robust), rbrv (Rank-Based Vote), zscore (Z-Score Consensus), softbrv (Soft Rank-Based Vote).

The Run DockM8 button at the bottom of the page will start the docking run. The log will be displayed on the page but also in your terminal window. If you ever want to stop the docking process you can use `Ctrl + C` to cancel the current process.

1.2 Command Line (via dockm8.py)

Please refer to Section 1.1 for a more detailed explanation of the various arguments.

Open a terminal (Ctrl + Alt + T on Ubuntu, or launch the Ubuntu app on Windows).

Activate the dockm8 python environment:

```
conda activate dockm8
```

Run the following command, replacing the *ARGUMENTS* with the arguments listed below as required:

```
python ~/DockM8/dockm8.py *ARGUMENTS*
```

You can also view the version with `--version` or get help with `--help`.

1.2.1 Required Arguments

`--software` (REQUIRED): The path to the software folder. In most cases this is where the DockM8 repository was downloaded to (usually `~/DockM8/software`).

`--receptor` (REQUIRED): The path to the protein file(s) (.pdb format). Multiple paths separated by spaces can be provided for ensemble mode. When providing multiple receptor files, the order should correspond to the order of `--reference_ligand` files.

`--docking_library` (REQUIRED): The path to the docking library file (.sdf format).

`--id_column` (REQUIRED): The unique identifier column used in the docking library.

`--docking_program` (REQUIRED): The docking program(s) to use. Must be one or more of:

- GNINA : GNINA molecular docking (CPU).
- GNINA_GPU : GNINA molecular docking (GPU-accelerated).
- SMINA : SMINA molecular docking.
- QVINA2 : QuickVina 2 molecular docking.
- QVINAW : QuickVina-W molecular docking.
- PLANTS : PLANTS molecular docking (requires separate license).

`--mode`: Choose mode with which to run DockM8. Options are:

- single (default): Regular docking on one receptor.
- ensemble : Ensemble docking on multiple receptor conformations.

`--ligand_protonation` (REQUIRED): The method to use for ligand protonation. Must be one of:

- GypsumDL : Use GypsumDL library to protonate library.
- None : Do not protonate library.

1.2.2 Global Settings

`--n_cpus`: The number of CPUs to use for the workflow. Default behavior is to use 90% of the available CPUs.

1.2.3 Pocket Finding Options

`--pocket_method`: The method to use for pocket determination. Must be one of:

- Reference (default): Uses a reference ligand to define pocket.
- RoG : Uses a reference ligand's radius of gyration to define pocket.
- Dogsitescorer : Call DogSiteScorer webserver to determine pocket coordinates.

`--reference_ligand`: The path to the reference ligand file(s) for pocket definition. Must be provided if using Reference or RoG pocket mode. Multiple files (separated by spaces) can be provided for ensemble mode, corresponding to the `--receptor` order.

`--manual_pocket`: A manual pocket can be defined by passing the center and size in the following format: `center:11.1,11.1,11.1`. This overrides `--pocket_method`.

`--dogsitescorer_method`: Choose which DogSiteScorer metric to use to select the binding site. Can be any one of Volume (default), Druggability_Score, Surface or Depth. The site with the highest value of the chosen metric will be used.

`--pocket_radius`: Radius for pocket finding in Angstroms. Default=10.

1.2.4 Protein Preparation Options

- prepare_protein: Whether or not protein files should be prepared (True or False). Default=True.
- protein_protonation: The method to use for protein protonation. Must be one of:
- protoss (default): Use the Protoss webserver.
 - pdbfixer : Use the PDBFixer library.
 - none : Skip protein protonation.
- fix_nonstandard_residues: Fix nonstandard residues during protein preparation (True or False). Default=True.
- fix_missing_residues: Fix missing residues during protein preparation (True or False). Default=True.
- remove_hetero: Remove heteroatoms during protein preparation (True or False). Default=True.
- remove_water: Remove water molecules during protein preparation (True or False). Default=True.

1.2.5 Ligand Preparation Options

- ligand_protonation (REQUIRED): The method to use for ligand protonation. Must be one of GypsumDL or None.
- conformers: The method to use for conformer generation. Must be one of:
- UFF : Use RDKit with the UFF forcefield to generate conformers.
 - MMFF (default): Use RDKit with the MMFF forcefield to generate conformers.
 - GypsumDL : Use GypsumDL to generate conformers.
- standardize_ids: Standardize compound IDs (True or False). Default=True. Setting to False may cause errors with numeric-only IDs or IDs containing special characters.
- remove_salts: Remove salts during ligand preparation (True or False). Default=True.
- min_ph: Minimum pH for ligand protonation when using GypsumDL. Default=6.4.
- max_ph: Maximum pH for ligand protonation when using GypsumDL. Default=8.4.
- pka_precision: pKa precision for ligand protonation when using GypsumDL. Default=1.0.

1.2.6 Docking Options

- n_poses: The number of poses to generate for each docking software. Default=10.
- exhaustiveness: The precision used if docking with SMINA/GNINA/GNINA_GPU/QVINA2/QVINAW. Default=8.
- bust_poses: Apply PoseBusters to filter out low-quality docking poses (True or False). Default=False. Note that enabling this may significantly increase the run time.

1.2.7 Pose Selection and Scoring Options

- selection_method: The method(s) to use for pose selection. Must be one or more of:
- bestpose (default): Takes the best pose from each docking program based on docking score.
 - Any rescoring function name (see --rescoring_functions below): DockM8 will select the best pose for each compound according to the specified scoring function.
- rescoring_functions: Which scoring functions to use for rescoring. Default: GNINA-Affinity CNN-Score. Must be one or more of:
- AD4
 - CHEMPLP
 - CNN-Affinity
 - CNN-Score
 - ConvexPLR
 - GenScore-balanced
 - GenScore-docking
 - GenScore-scoring
 - GNINA-Affinity
 - GNINA-All (returns 3 columns: GNINA-Affinity, CNN-Score, CNN-Affinity)

- KORP-PL
- LinF9
- NNScore
- PLP
- RFScoreVS
- RTMScore
- SCORCH
- Vinardo

--consensus_method: Which consensus method to use. Default: rbr. Must be one of:

- ecr : Element Count Rank
- rbr : Rank-Based Robust
- rbv : Rank-Based Vote
- zscore : Z-Score Consensus
- softrbv : Soft Rank-Based Vote

--ensemble_top_percent: Report compounds in the top X percent across all receptor structures when using ensemble mode. Default=1.0.

1.2.8 Decoy Generation Options

--generate_decoys: Whether or not to generate decoys for performance assessment (True or False). Default=False.

--actives: The path to the file containing active compounds (.sdf format). Compounds do not need to be labelled, DockM8 will take care of the labelling.

--decoy_model: Choose which decoy model to use for decoy generation. Can be one of: DUDE (default), DEKOIS, DUDE_P.

--n_decoys: Number of decoys to generate per active compound. Default=20.

--performance_thresholds: Thresholds (as percentages) for performance metrics calculation. Default: 1 2 5.

1.2.9 Example Command

The following is a typical DockM8 command to illustrate the above arguments:

```
python ~/DockM8/dockm8.py --mode single --software ~/DockM8/software --receptor
→ ~/path/to/receptor.pdb --pocket_method Reference --reference_ligand
→ ~/path/to/reference.sdf --docking_library ~/path/to/docking_library.sdf
→ --id_column ID --prepare_protein True --protein_protonation protoss
→ --conformers MMFF --ligand_protonation GypsumDL --docking_program GNINA SMINA
→ PLANTS --n_poses 10 --exhaustiveness 8 --bust_poses False --selection_method
→ bestpose GNINA-Affinity --rescoring_functions GNINA-Affinity ConvexPLR AD4
→ NNScore --consensus_method rbr
```

1.3 Jupyter Notebooks

DockM8 provides three Jupyter Notebooks for different workflows:

- dockm8.ipynb : Standard single-receptor virtual screening workflow. Follow the step-by-step instructions in the notebook to dock a compound library to a single receptor.
- dockm8_ensemble.ipynb : Multi-receptor ensemble docking workflow. Dock compounds against multiple receptor conformations and identify consensus hits.
- dockm8_decoys.ipynb : Two-phase decoy optimization and parameter tuning workflow. Generate decoys, evaluate performance, and identify optimal parameters for your target.

All notebooks are located in the DockM8 root folder. Follow the guidance from Sections 1.1 and 1.2 as well as the instructions in the notebooks themselves to run the workflows.

1.4 Analysis Pipeline

DockM8 includes an analysis pipeline in the analysis/ directory for reproducing the benchmark figures from the publication. This pipeline supports the DEKOIS, DUD-E, and Lit-PCBA benchmark datasets. For detailed instructions on downloading the data and running the analysis, please refer to analysis/README.md in the DockM8 repository.

2 Frequently Asked Questions

If GNINA does not run, you may need to run the following command to point GNINA to the lib folder in your conda installation directory:

```
export LD_LIBRARY_PATH=$LD_LIBRARY_PATH:**PATH_TO**/miniconda3/envs/dockm8/lib/
```

Replace ****PATH_TO**** with the actual path to your Miniconda or Anaconda installation (e.g., `~/miniconda3/envs/dockm8/lib/`).

3 Citation

For more information about the performance of DockM8 or for bibliographic references to the various software used in the workflow, please see the relevant publication or visit our [website](#). A machine-readable citation file (`CITATION.cff`) is included in the DockM8 repository.